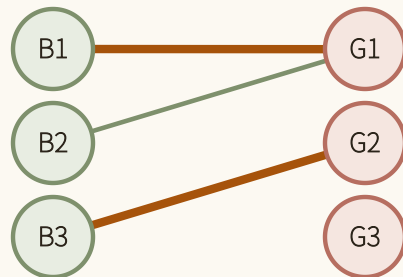


From the Equality Graph to the Project Matrix

The same Hungarian algorithm, stored in a different form

Equality graph G_h



Reduced-cost matrix

	G1	G2	G3
B1	0	6	16
B2	0	4	4
B3	23	0	8

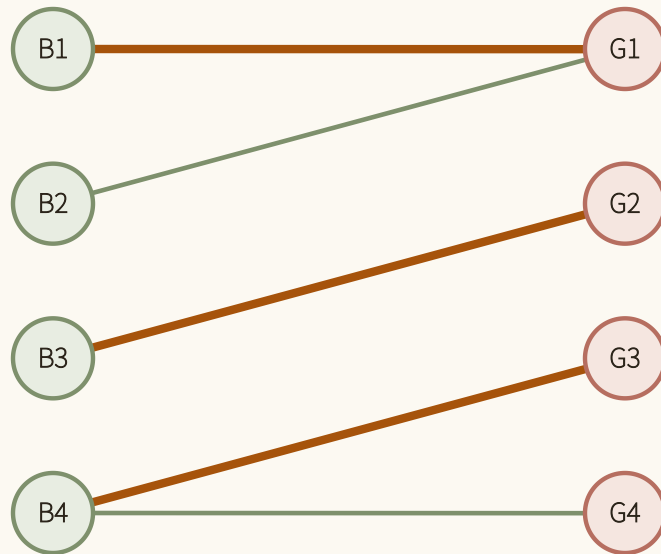
graph idea $\rightarrow$ matrix operation

One graph object becomes one matrix object

Read every arrow from left to right: Part I terminology → project terminology.

Part I: equality graph G_h

tight edges only



edge weight $w(b_i, g_j)$
 slack $h(b_i) + h(g_j) - w$
 equality edge in G_h
 matching M



Project: reduced-cost matrix

tight edge $\Leftrightarrow$ zero cell

	G1	G2	G3	G4
B1	0	6	482	16
B2	0	478	4	4
B3	23	0	8	483
B4	484	12	0	0



original cell $LS[i][j]$
 reduced cost $R[i][j]$
 zero cell
 independent zeros (IZ)

Project preprocessing creates the slack matrix

$N = 4$, $DB = -100N = -400$, and $HLS = H = 88$.

1 • Original LS matrix

$H = 88$

	G1	G2	G3	G4
B1	82	76	DB	61
B2	78	DB	74	69
B3	65	88	80	DB
B4	DB	72	84	79



2 • Cost = $H - LS$

every DB becomes 488

	G1	G2	G3	G4
B1	6	12	488	27
B2	10	488	14	19
B3	23	0	8	488
B4	488	16	4	9



3 • Row + column reduction

row min [6,10,0,4] • col min [0,0,0,5]

	G1	G2	G3	G4
B1	0	6	482	16
B2	0	478	4	4
B3	23	0	8	483
B4	484	12	0	0

Connection to Part I

Row reduction gives $h(b) = \text{row maximum}$.

Column reduction is optional safe tightening.

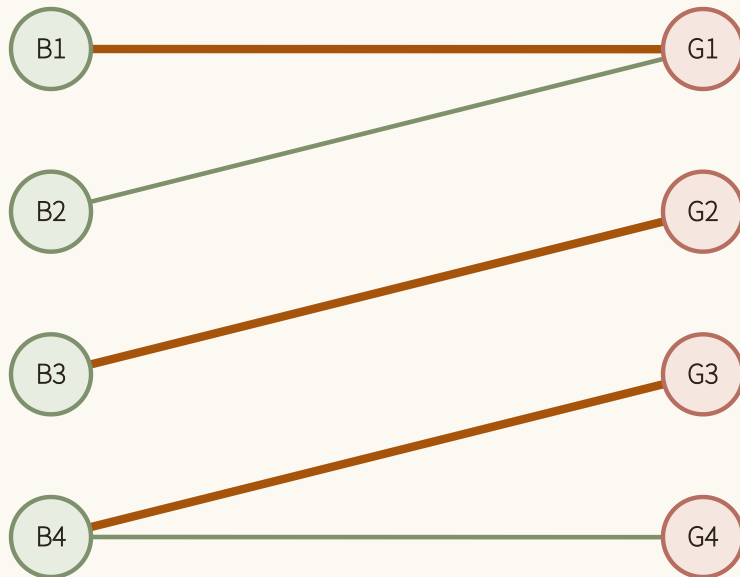
After reduction: $R[i][j] = h(b_i) + h(g_j) - w(b_i, g_j)$. Therefore zero $\Leftrightarrow$ equality edge.

A zero matching is the matching in G_h

The project finds the maximum possible set of independent zeros.

Equality graph G_h

B2 is still unmatched



Independent zeros (IZ)

maximumZeroMatching(...)

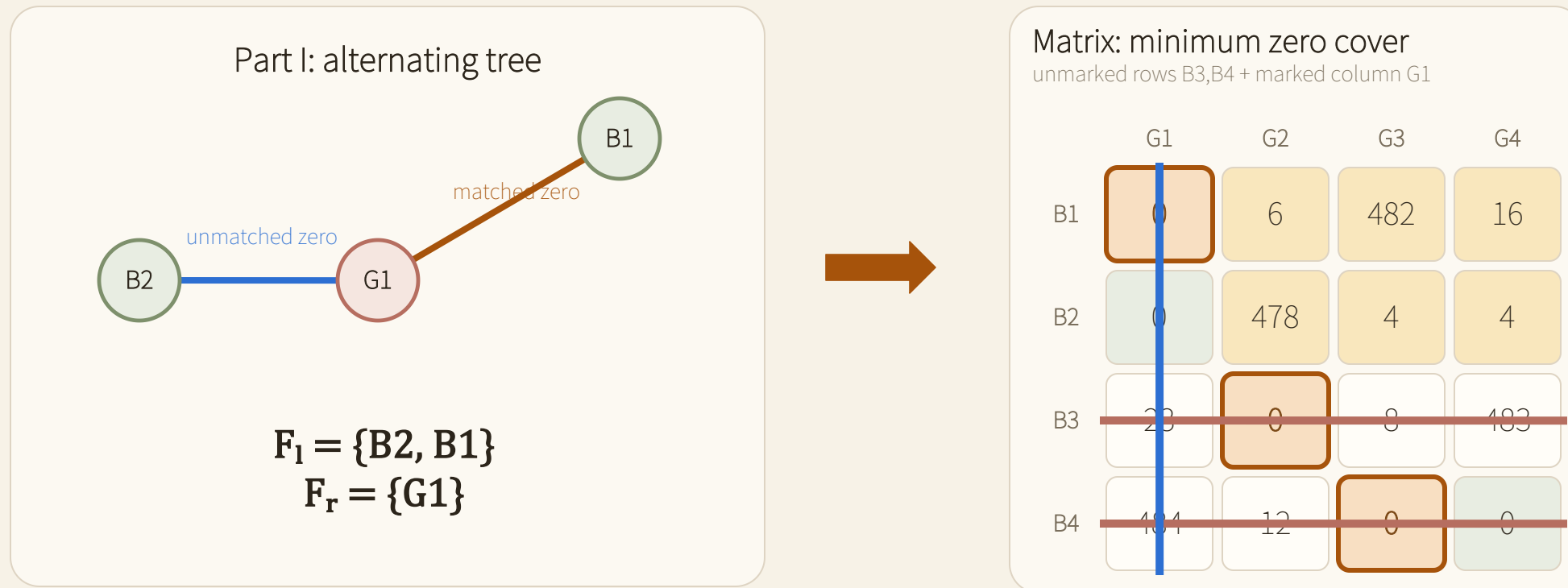
	G1	G2	G3	G4
B1	0	6	482	16
B2	0	478	4	4
B3	23	0	8	483
B4	484	12	0	0

$$IZ = 3 < N = 4$$

No perfect equality matching yet → build the alternating forest.

The alternating forest becomes matrix marks and cover lines

F_l and F_r are computed first; the minimum zero cover follows from them.



Uncovered cells = $F_l \times (R - F_r)$, exactly the domain of the δ formula.

The smallest uncovered cost is the same δ

Part I updates labels; the matrix updates the corresponding edge slacks.

Part I label update

$$\delta = \min_{b \in F_l, g \notin F_r} [h(b) + h(g) - w(b, g)]$$

for $b \in F_l$: $h(b) -= \delta$
for $g \in F_r$: $h(g) += \delta$



Before: $\delta = 4$

minimum uncovered cost

	G1	G2	G3	G4
B1	0	6	482	16
B2	0	478	4	4
B3	23	0	8	483
B4	484	12	0	0



After adjustment

uncovered -4 • double-covered $+4$

	G1	G2	G3	G4
B1	0	2	478	12
B2	0	474	0	0
B3	27	0	8	483
B4	488	12	0	0

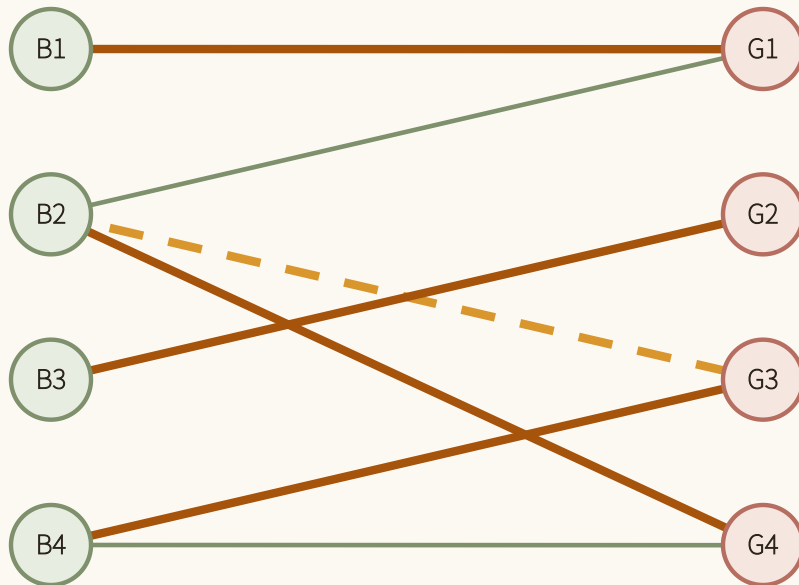
Matrix rule: uncovered $-\delta$, double-covered $+\delta$, one line unchanged. New zeros enter G_h .

A new equality edge completes the assignment

The project finds four independent zeros, then maps their positions back to the original LS matrix.

Perfect matching in G_h

project zero-matching order



Decode positions in original LS

B1-G1, B2-G4, B3-G2, B4-G3

	G1	G2	G3	G4
B1	82	76	DB	61
B2	78	DB	74	69
B3	65	88	80	DB
B4	DB	72	84	79

$$82 + 69 + 88 + 84 = 323 \quad \text{and} \quad \Sigma h = 323 \Rightarrow \text{optimal.}$$

Deal Breaker is a large penalty, then post-processing

DB keeps the square assignment problem complete, but makes every valid pair preferable.

Project rule

$$\text{DB} = -100N$$

For $N = 4$:

$$\text{DB} = -400$$

$$\text{Cost}(\text{DB}) = 88 - (-400) = 488$$

One fewer DB is always worth more than any possible change in the other $N-1$ scores.



If one DB is unavoidable

$N = 3$, $\text{DB} = -300$

	G1	G2	G3
B1	90	DB	DB
B2	80	DB	DB
B3	DB	70	60

Complete assignment

$$\text{B1-G1} = 90$$

$$\text{B3-G2} = 70$$

$$\text{B2-G3} = \text{DB}$$

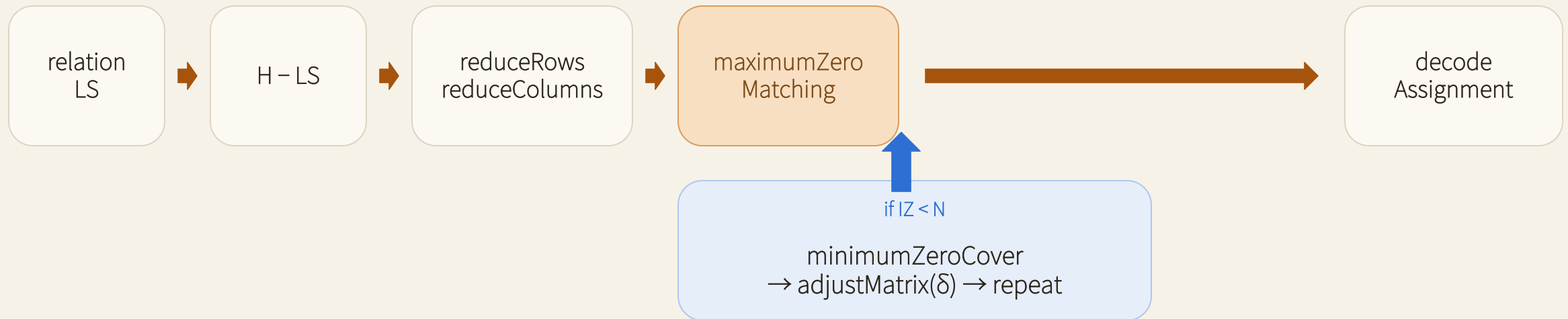
Post-processing

remove DB pair
B2 and G3 → unmatched
TS = 160

Optimization order: maximize valid pair count first; among those assignments, maximize total score.

The project repeats the exact graph algorithm in matrix form

Only the representation changes; the mathematical decisions stay the same.



Part I graph

equality edge in G_h
 matching + augmenting path
 F_l and F_r
 δ = minimum slack
 perfect matching in G_h

Project matrix

$\text{matrix}[\text{row}][\text{col}] == 0$
 independent zeros + augment(...)
 marked rows and marked columns
 minimum uncovered cost
 N independent zeros